

Owen Karmel

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Education

Cornell University, Ithaca NY

Aug, 2023 - May, 2027

- Bachelor of Arts pursuing majors in Biological Sciences and Physics
- Current GPA: 3.85

Research Experience

Research Assistant, Biomedical Engineering, Jiang Lab, Cornell University

Aug, 2024 - Current

- Developing mRNA-based therapeutic approaches that selectively target senescent cells as a novel strategy to address aging and age-related diseases
- Leading two independent projects to (1) investigate the potential geroprotective effects of miRNA sponges, and (2) design a novel self-amplifying mRNA platform.
 - Designed and tested eight unique miRNA-responsive fluorescent reporters and then compared both 5' vs 3' mediated silencing mechanisms.
 - Optimized a series of low-cost stem-loop mediated RT-qPCR assays to accurately measure the absolute abundance of senescence-associated miRNAs across three different cell types *in vitro*.
 - Purified extracellular vesicles with size exclusion chromatography and characterized them via nanoparticle tracking analysis (NTA) and direct light scattering (DLS).
 - Investigated via flow cytometry the integration of a riboswitch into self-amplifying RNA for prolonged, non immunogenic expression
- Supported a diagnostic project utilizing Raman spectroscopy and machine learning to identify senescent cells and detect metabolic changes in a rapid, label-free manner.

Research Intern, Thymune Therapeutics

June, 2026 - Current

- Optimized generation of thymus cells and developed for their functional characterization
- Built internal website for effective scRNA-seq data visualization with Azure DevOps

Engineering Intern, Regeneron Pharmaceuticals

May, 2024 - Aug 2024

- Conducted 3-month accelerated aging study of pre-filled syringe shelf life of 160 device samples. Discovered via ANOVA significant increase in plunger depression force and proposed a mechanistic explanation.
- Presented results to department leadership and wrote a 15-page technical report.

Research Assistant, Biomedical Engineering, Cosgrove Lab, Cornell University

Dec, 2023 - May 2024

- Developed a FACS protocol for isolating primary macrophages using surface markers identified through scRNA-seq analysis

Honors & Awards

Cornell Rawlings Presidential Research Fellowship (2025)

- Top 2% of Cornell undergraduates demonstrating strong commitment to research with \$7000 research and conference travel support account

Jiang Fellow (2025)

- Highly selective \$11000 award supporting Cornell students to perform summer research at a startup company of their choosing

Cornell Summer Experience Grant (2025)

- Competitive \$5000 award for pursuing summer research

Cornell Einhorn Center Engaged Student Grant (2024)

- \$1000 award for commitment to advocacy/outreach, e.g. longevity research advocacy

Service & Leadership

Volunteer, Longevity Biotechnology Association

Nov, 2024 - Current

- Wrote report evaluating 44 relevant companies while volunteering at the national Longevity Biotech Association to promote membership, outreach and investment in the longevity field
- Photography and social media lead for October 23 biotech pitch event in NYC

Teacher, Splash! at Cornell

Dec, 2025 - Current

- Taught hands on classes in molecular biology and rocketry to middle and high schoolers

Co-founder and President, Cornell Longevity Association

Dec, 2025 - Current

- Independently organized speaker events with biotech CEOs and a director of geriatric medicine, each with 20-60+ attendees

Editor, Cornell Undergraduate Research Journal

Aug, 2025 - Current

- Peer review of over 20 undergraduate research paper submissions in all disciplines

Peer Mentor, Cornell University

Oct, 2024 - Current

- Served as a mentor for 6 undergraduates in professional development and finding research opportunities as member of Peer Mentorship Program
- Led 14 mile summit of Algonquin and Cascade mountains in Adirondacks in NOLS program

Presentations

Karmel, O. et al. (2026, October 24). *Tetracycline-responsive aptazymes enable tunable expression of self-amplifying RNA* [Presentation]. Biomedical Engineering Society Annual Meeting, Orlando, FL, United States.

Karmel, O., & Beaman, H. (2026, August 12). *scRNA-seq visualization website* [Presentation]. Thymune Company-Wide Meeting, Boston, MA, United States.

Karmel, O. (2024, August 7). *Characterization of a novel accelerated aging strategy for assessing device functional stability* [Presentation and technical report]. Combination Product Dept. Leadership at Regeneron, Albany, NY, United States.

Wet Lab Skills: Mammalian cell culture (primary human dermal fibroblasts, NIH3T3s, HEK293s), restriction enzyme and Gibson cloning, flow cytometry (live dead, gene expression, SA- β -gal assay), in vitro synthesis of mRNA, transfections, RT-qPCR (for both mRNA and miRNA), western blotting

Computational Skills: Benchling for molecular biology applications (mRNA design, sequence alignment, and plasmid assembly). Python packages (NumPy, Pandas, scikit-learn, Torch), Affinity Designer and Graphpad Prism for scientific graphics, R and JMP for statistical analysis.

Coursework: Honors Organic Chemistry I, Stem Cells and Regeneration, Introduction to Machine Learning, Computational Genetics and Genomics, Anatomy and Physiology, Quantum Mechanics